Northwestern University
Dept. of Computer Science

Tommy McMichen

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Research Interests

My interests lie in *compilers*, specifically how we can leverage modern programming languages to design general-purpose *intermediate representations* that (1) improve the precision of *static analysis* by preserving high-level information, (2) enable novel *optimizations* on the layout and organization of memory, and (3) open the door to *compiler-runtime codesigns* that rethink existing hardware abstractions.

Education

Northwestern University, Evanston, Illinois, USA — 2020 – Present
Ph.D. in Computer Science, *Advised by Simone Campanoni* — (Expected 2026)
M.Sc. in Computer Science, *Advised by Simone Campanoni* — 2023

Rose-Hulman Institute of Technology, Terre Haute, Indiana, USA — 2016–2020
B.Sc. in Computer Engineering and Computer Science

Publications

Automatic Data Enumeration for Fast Data Collections, *CGO 2026.*
Tommy McMichen, Simone Campanoni.

Saving Energy with Per-Variable Bitwidth Speculation, *ASPLOS 2025.*
Tommy McMichen, David Dlott, Panitan Wongse-ammat, Nathan Greiner, Hussain Khajanchi, Russ Joseph, Simone Campanoni.

Representing Data Collections in an SSA Form, *CGO 2024.*
Tommy McMichen, Nathan Greiner, Peter Zhong, Federico Sossai, Atmn Patel, Simone Campanoni.

Getting a Handle on Unmanaged Memory, *ASPLOS 2024.*
Nick Wanninger, **Tommy McMichen**, Simone Campanoni, Peter Dinda.

Program State Element Characterization, *CGO 2023.*
Enrico Armenio Deiana, Brian Suchy, Michael Wilkins, Brian Homerding, **Tommy McMichen**, Katarzyna Dunajewski, Peter Dinda, Nikos Hardavellas, Simone Campanoni.

NOELLE Offers Empowering LLVM Extensions, *CGO 2022.*
Angelo Matni, Enrico Armenio Deiana, Yian Su, Lukas Gross, Souradip Ghosh, Sotiris Apostolakis, Ziyang Xu, Zujun Tan, Ishita Chaturvedi, Brian Homerding, **Tommy McMichen**, David I. August, Simone Campanoni.

Fine-Grained Acceleration using Runtime Integrated Custom Execution (RICE), *CASES 2019.*
Leela Pakanati, **Tommy McMichen**, Zachary Estrada.

Invited Talks

"Representing Data Collections for Analysis and Transformation"
Languages, Systems, and Data Seminar, *University of California, Santa Cruz.* — October 2025
Computer Architecture Group Meeting, *University of Cambridge.* — March 2024
Tech Talk, *Rose-Hulman Institute of Technology.* — October 2023
Student Seminar Series, *Northwestern University.* — October 2023
Constellation Workshop, *Northwestern University.* — July 2023

"Towards Collection-Oriented Compilation in LLVM"
LLVM Developers' Meeting, *Santa Clara, California.* — October 2025

Industry Experience

Meta, Menlo Park, California, USA — Summer 2025
Software Engineering Intern, Programming Languages and Runtimes, Android Native Compiler Team
- Improved the representation of ClangIR, an open-source C/C++ MLIR compiler.
- Developed analyses and transformations for C++ move semantics in ClangIR.
- Lead ClangIR open-source development efforts through issue creation and code reviews.

Texas Instruments, Dallas, Texas, USA — Summer 2019, Summer 2020
Digital Design Engineering Intern, Embedded Processors, Analytics Team
- Performed integration testing for hardware implementation of cache coherence protocol.
- Developed coverage metrics for cache coherence testing.
- Implemented automatic generation of RTL and TLM from descriptor files.

National Instruments, Austin, Texas, USA — Summer 2018
R&D Software Engineering Intern, Digitizers
- Designed and implemented FPGA logic for new function generator feature with LabVIEW.
- Added kernel, driver and API support for new function generator feature.
- Implemented full driver stack support for highly-customisable oscilloscope triggers.
- Communicated with multiple teams to add new .NET API entry points.

Teaching Experience

Teaching Assistant, *COMP_SCI 322 Compiler Construction*, Prof. Simone Campanoni. Winter 2022
Resident Tutor, *Computer Science and Computer Engineering Departments.* Aug. 2019 – May 2020

Service

Artifact Evaluation Committee, International Conference on Compiler Construction (CC).	2026
Board Member, Computer Science Social Initiative, Northwestern University.	2021
Member, CS Ph.D. Orientation Planning Committee, Northwestern University.	2022
Member, CS Ph.D. Visit Day Planning Committee, Northwestern University.	2022
Student Volunteer, International Symposium on Microarchitecture (MICRO).	October 2022
Chairperson, IEEE, Rose-Hulman Institute of Technology student branch.	August 2019
Corresponding Secretary, Eta Kappa Nu (HKN), Epsilon Eta Chapter.	August 2019
Member, Eta Kappa Nu (HKN), Epsilon Eta Chapter.	May 2018

Funding and Awards

LLVM Foundation Student Travel Grant, *LLVM Developers' Meeting.*	2025
NSF Student Travel Grant, *ASPLOS.*	2025
NSF Student Travel Grant, *HPCA/PPoPP/CGO.*	2024
NSF Student Travel Grant, *HPCA/PPoPP/CGO.*	2023
IP/ROP Student Travel Award.	2019
NSF Student Travel Grant, *ESweek.*	2019
IP/ROP Student Project Grant.	2018

Research Advising

Akash Deo, M.Sc., *Designing compiler tools for AI-assisted vectorization.*	2025 – Present
Benjamin Ye, M.Sc., *Characterizing differences between LLVM front-ends.*	2025 – Present
Benjamin Ye, B.Sc., *Automatically generating MemOIR from Rust.*	2024 – 2025